TODO APP

Task Manager with Priority & Filter System

PROJECT DOCUMENTATION

Version 1.0 · 2024 · Yohannes Bayeh

Field	Detail
Project Name	Todo App
Type	Frontend SPA — Personal/Portfolio Project
Developer	Yohannes Bayeh
Year	2024
Live URL	https://todoappnew-six.vercel.app/
GitHub	https://github.com/John-bayeh/Todoappnew
Stack	React 19 + Vite + CSS3 (No backend, no database)
Theme	Dark Purple — #1E1B4B background, #7C3AED accent

SECTION 1 — PROJECT OVERVIEW

1.1 Executive Summary

Todo App is a clean, fast, and intuitive task management Single Page Application (SPA) built with React 19 and Vite. It allows users to create, organize, prioritize, and filter tasks entirely in the browser — with no backend, no database, and no account required. The application features a dark purple UI theme with gradient accents, demonstrating advanced React patterns including useState, useMemo, component architecture, and polished visual design.

1.2 Problem Statement

Users needed a distraction-free task manager that works instantly without registration friction, backend dependencies, or complex setup. Existing tools are often overloaded with features that slow down simple task tracking.

1.3 Solution

A pure frontend React SPA that runs entirely in the browser with zero signup, zero backend calls, and zero latency. Tasks are managed in React state and rendered instantly on every interaction. The dark purple gradient design creates a focused, distraction-free environment for task management.

1.4 Key Design Decisions

Decision	Reason
No backend / no database	Demonstrates pure React state management without infrastructure complexity
useState only (no Redux)	App complexity does not require global state — keeps architecture clean
useMemo for filtering	Avoids unnecessary re-computation on every render
Vite over CRA	Faster HMR, smaller bundle, modern build tooling
Dark purple theme	Creates focused, distraction-free task management environment
No auth / no login	Zero friction — demonstrates UX decision-making

SECTION 2 — FEATURES & FUNCTIONALITY

2.1 Core Features

Feature	Description	Implementation
Task Creation Form	Add tasks with name, description, due date, priority	Controlled form with useState
Filter by Status	All / Active / Completed dropdown filter	useMemo client-side filter
Sort Tasks	Sort By dropdown — due date or priority	Array sort on Date objects / priority weight
One-Click Status Toggle	Mark task complete/incomplete instantly	setTasks state update
Priority Selection	Low / Medium / High priority dropdown	Conditional CSS classes per priority
Due Date Tracking	Date picker with visual urgency indicator	Date comparison logic
Task Deletion	Remove tasks from list	Filter out by task ID
Empty State	No tasks found message when list is empty	Conditional render

2.2 Priority System

Priority Level	Color	Meaning
High	Red / Urgent	Must complete soon
Medium	Yellow / Warning	Important but not urgent
Low	Muted gray	Nice to have

2.3 Filter & Sort Logic

All filtering and sorting happens client-side using React's useMemo hook:

- All — show every task regardless of completion status
- Active — show only tasks where task.completed === false
- Completed — show only tasks where task.completed === true

- Sort by Due Date — ascending: new Date(a.dueDate) - new Date(b.dueDate)
- Sort by Priority — weight map: High=3, Medium=2, Low=1 (descending)

SECTION 3 — TECHNICAL ARCHITECTURE

3.1 Technology Stack

Layer	Technology	Purpose
Framework	React 19	Component-based UI with hooks
Build Tool	Vite	Fast HMR, optimized production build
Language	JavaScript (ES6+)	No TypeScript — clean JS
Styling	CSS3 + Gradients	Dark purple theme with gradient accents
State Management	useState + useMemo	Local component state
Deployment	Vercel	Static SPA hosting with global CDN
Backend	None	Pure frontend — no server required
Database	None	useState only — no persistence

3.2 Component Architecture

Component	Responsibility
App.jsx	Root — holds all task state, filter state, sort state
TaskForm	Controlled form for creating tasks with validation
TaskList	Renders filtered and sorted task cards
TaskCard	Individual task with priority, due date, toggle, delete
FilterBar	Status filter dropdown (All/Active/Completed)
SortControls	Sort By dropdown (Due Date/Priority)

3.3 State Management

State	Type	Purpose

tasks	Array	Master list of all task objects
filter	String	'All' | 'Active' | 'Completed'
sortBy	String	'dueDate' | 'priority'
filteredTasks	useMemo	Derived from tasks + filter + sortBy

3.4 Data Model

Field	Type	Description
id	String (UUID)	Unique task identifier
title	String	Task name — required
description	String	Optional task details
dueDate	String (ISO date)	Deadline for sorting and urgency
priority	String	'High' | 'Medium' | 'Low'
completed	Boolean	Task completion status
createdAt	String (ISO date)	Task creation timestamp

SECTION 4 — SCOPE & REQUIREMENTS

4.1 Functional Requirements

ID	Requirement	Priority	Status
FR-01	Create task with name, description, due date, priority	High	✅ Done
FR-02	Filter tasks by All / Active / Completed	High	✅ Done
FR-03	Sort tasks by due date	High	✅ Done
FR-04	Sort tasks by priority	High	✅ Done
FR-05	Toggle task complete/incomplete	High	✅ Done
FR-06	Delete task from list	High	✅ Done
FR-07	Color-coded priority indicators	Medium	✅ Done
FR-08	Due date urgency indicator	Medium	✅ Done
FR-09	Empty state message	Low	✅ Done
FR-10	Instant filter/sort with no API calls	High	✅ Done

4.2 Non-Functional Requirements

ID	Requirement	Target
NFR-01	Filter/sort response time	Instant — <16ms
NFR-02	Page load time	<2 seconds
NFR-03	No backend dependency	Pure frontend SPA
NFR-04	Mobile responsive	All screen sizes
NFR-05	Lighthouse performance	95+

SECTION 5 — DESIGN

5.1 Design System

Element	Value	Usage
Background	#1E1B4B (Dark Navy Purple)	Page background
Gradient	Purple to Dark gradient	Full page background gradient
Accent	#7C3AED (Purple)	Buttons, Add Task, active states
Light Accent	#A78BFA (Light Purple)	Highlights, labels
Form Background	#2D2A5E	Task creation card background
Input Border	#7C3AED	Form input borders
Text Primary	#FFFFFF	Headings and labels
Text Muted	#9CA3AF	Secondary text
High Priority	#EF4444 (Red)	High priority badge
Medium Priority	#F59E0B (Amber)	Medium priority badge
Low Priority	#6B7280 (Gray)	Low priority badge

5.2 UI Layout

- Centered card layout on dark purple gradient background
- Filter (All/Active/Completed) and Sort dropdowns at the top
- Add New Task card below with all form fields
- Task list renders below the form
- Empty state 'No tasks found.' shown when list is empty

SECTION 6 — TESTING

6.1 Manual Feature Testing

Test Case	Expected Result	Result
Create task with all fields	Task appears in list	✅ Pass
Create task with no title	Validation — title required	✅ Pass
Toggle task complete	Task moves to Completed state	✅ Pass
Filter: Active	Only incomplete tasks shown	✅ Pass
Filter: Completed	Only completed tasks shown	✅ Pass
Filter: All	All tasks shown	✅ Pass
Sort by Due Date	Earliest deadline first	✅ Pass
Sort by Priority	High → Medium → Low	✅ Pass
Delete task	Task removed instantly	✅ Pass
Empty state	'No tasks found.' shown	✅ Pass

6.2 Responsive Testing

Device	Screen Size	Result
iPhone 12 Pro	390px	✅ Pass
iPad Air	820px	✅ Pass
Desktop	1440px	✅ Pass

6.3 Performance

Metric	Target	Result
Lighthouse Performance	95+	✅ 97
Page Load Time	<2 seconds	✅ 1.2 seconds
Filter/Sort Speed	Instant	✅ <8ms

6.4 Limitations (By Design)

- No data persistence — refreshing clears all tasks (useState only)
- No multi-device sync — tasks exist in current browser session only
- No collaboration — single user only

These are intentional design constraints — the goal was to demonstrate React state management patterns, not build a full production task manager.

SECTION 7 — DEPLOYMENT

7.1 Deployment

Component	Platform	Detail
Frontend SPA	Vercel	https://todoappnew-six.vercel.app/
Build Command	npm run build	Vite production build → dist/
Output Directory	dist	Static files on Vercel CDN
Backend	None	No server needed
GitHub	John-bayeh/Todoappnew	Source code repository

7.2 What This Project Demonstrates

- React hooks mastery — useState, useMemo
- Component architecture — clean separation of concerns
- UX decision-making — zero friction, instant interactions
- CSS design — dark purple gradient theme, priority color system
- Performance awareness — useMemo for derived state
- Vite build tooling — modern frontend setup

APPENDIX — DOCUMENT CONTROL

Version	Date	Author	Changes
1.0	2024	Yohannes Bayeh	Initial documentation — complete